Lecture 01, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

Differential Equation

Welcome!

Differential equation: an equation where one or more derivatives of the unknown function appear.

Example: $PV = nRT$ NOT a diff. eq. P, V, and T can be unknown, but no derivative appear.

Example: $\frac{\mathrm{d}v}{\mathrm{d}t} = Mg - \gamma v$ where $v(t)$ is the unknown, IS a diff. eq.

Ordinary and Partial Differential Equations

ODE: Ordinary differential equation, the unknown function depends only on one independent variable.

Example: $\frac{dP}{dt} = P - t^2$ where $P(t)$ is a function of t

PDE: Partial differential equation, the unknown function depends on more than one ind. variable.

Example: $\frac{\partial \mu}{\partial t} = k\frac{\partial^2 \mu}{\partial t^2}$ where $\mu(x, t)$ is a function of x and t

timestamp: 02:20

System of Differential Equations

System of differential equations: multiple unknown functions and multiple coupled equations.

Ex:

$$\begin{cases} \frac{\mathrm{d}u}{\mathrm{d}t} = 3u - uv \\ \frac{\mathrm{d}v}{\mathrm{d}t} = u - v \end{cases}$$

timestamp: 04:34

Order of a Differential Equation

Order of an ODE: highest derivative

Ex: $y'' + x^3(y')^4 = x^5$ is 2nd order

Ex: $(y')^5 + y'''y'' + x^2y^2 = y^6$ is 3rd order

timestamp: 05:55

Linear and Nonlinear Differential Equations

Linear: unknown function and its derivative appear linearly (i.e. at power 1 or 0, not multiplied to each other, not inside nonlinear functions)

Nonlinear: otherwise

Ex: $y'' + x^2 y' + y - 5 = 0$ is linear

Ex: $y'' + yy' = x^2$ is nonlinear (because of yy')

Ex: $y'' + y^{\frac{1}{2}} = 0$ is nonlinear (because of $y^{\frac{1}{2}}$)

timestamp: 09:05

Linearization

Ex: $y'' + \sin(y) = 0$ is nonlinear because $\sin(y) \approx y - \frac{y^3}{3!} + \frac{y^5}{5!}$

by taylor series expansion so it is a nonlinear function

Linearization: Ex, for small y we can approximate $\sin(y) \sim y$ so

that we get the approximate equation $y'' + y = 0$, called the

linearization

Homogeneous and Inhomogeneous Equations

Homogeneous equation: all the terms include the unknown function or one of its derivatives.

Inhomogeneous equation: otherwise.

Ex: $y'' + x^2 y' + y - 5 = 0$ is inhomogeneous (and linear). Rewrite as

$y'' + x^2 y' + y = 5$

$y'' + x^2 y + y = 0$ is homogeneous

Ex: $x^2 y' = y'' - y$ is homogeneous. Rewrite as $x^2 y' - y'' + y = 0$

timestamp: 13:00

To solve an inhomogeneous equation we will need to solve the homogeneous equation first.

Ex: $y'' + x = y' + y$ is inhomogeneous. Rewrite as $y'' - y' - y = -x$

We will first solve the equation $y'' - y' - y = 0$

timestamp: 15:50

Solution of a differential equation

A general n-th order ODE can be written as

$$F(x, y, y', y'', \cdots, y^{(n)}) = 0 \qquad (*)$$

The standard notation is $y^{(n)} = G(x, y, y', \cdots, y^{(n-1)})$

The function $y(x)$ is a solution to (*) in an interval I if

$y, y', \cdots, y^{(n)}$ exist in I and they satisfy (*) $\forall x \in I$ (for every x in I)

Ex: $y'' - y' - 2y = 0$

The function $y(x) = \mathrm{e}^{2x}$ is a solution $\forall x \in \mathbb{R}$ (the set of all real numbers) because $y, y' = 2\mathrm{e}^{2x}$, and $y'' = 4\mathrm{e}^{2x}$ exist $\forall x \in \mathbb{R}$ and they satisfy the equation $\forall x \in \mathbb{R}$:

$$4\mathrm{e}^{2x} - 2\mathrm{e}^{2x} - 2\mathrm{e}^{2x} = 0$$

3 common equations

The following equations for $y(x)$ are very common:

$$y' = ay \tag{1}$$

$$y' + a^2 y = 0 \tag{2}$$

$$y' - a^2 y = 0 \tag{3}$$

where $a \in \mathbb{R}$ is a constant.

timestamp: 26:08

Find the corresponding solutions (these are not in the right order)

$$y(x) = A\mathrm{e}^{ax} + B\mathrm{e}^{-ax} \qquad \text{(i)}$$

$$y(x) = A\mathrm{e}^{ax} \qquad \text{(ii)}$$

$$y(x) = A\cos(ax) + B\sin(ax) \qquad \text{(iii)}$$

where A and B are free parameters.

timestamp: 26:45

Simple Population Model

$P(t)$ population of fish in a pond.

Set the rate of change equation as

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \alpha P - H$$

where α is the rate of growth due to births (proportional to P) and

H is a constant due to harvesting (we harvest a fixed amount per

unit time)

timestamp: 28:44

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 2P - 4 = 2(P - 2)$ in some units.

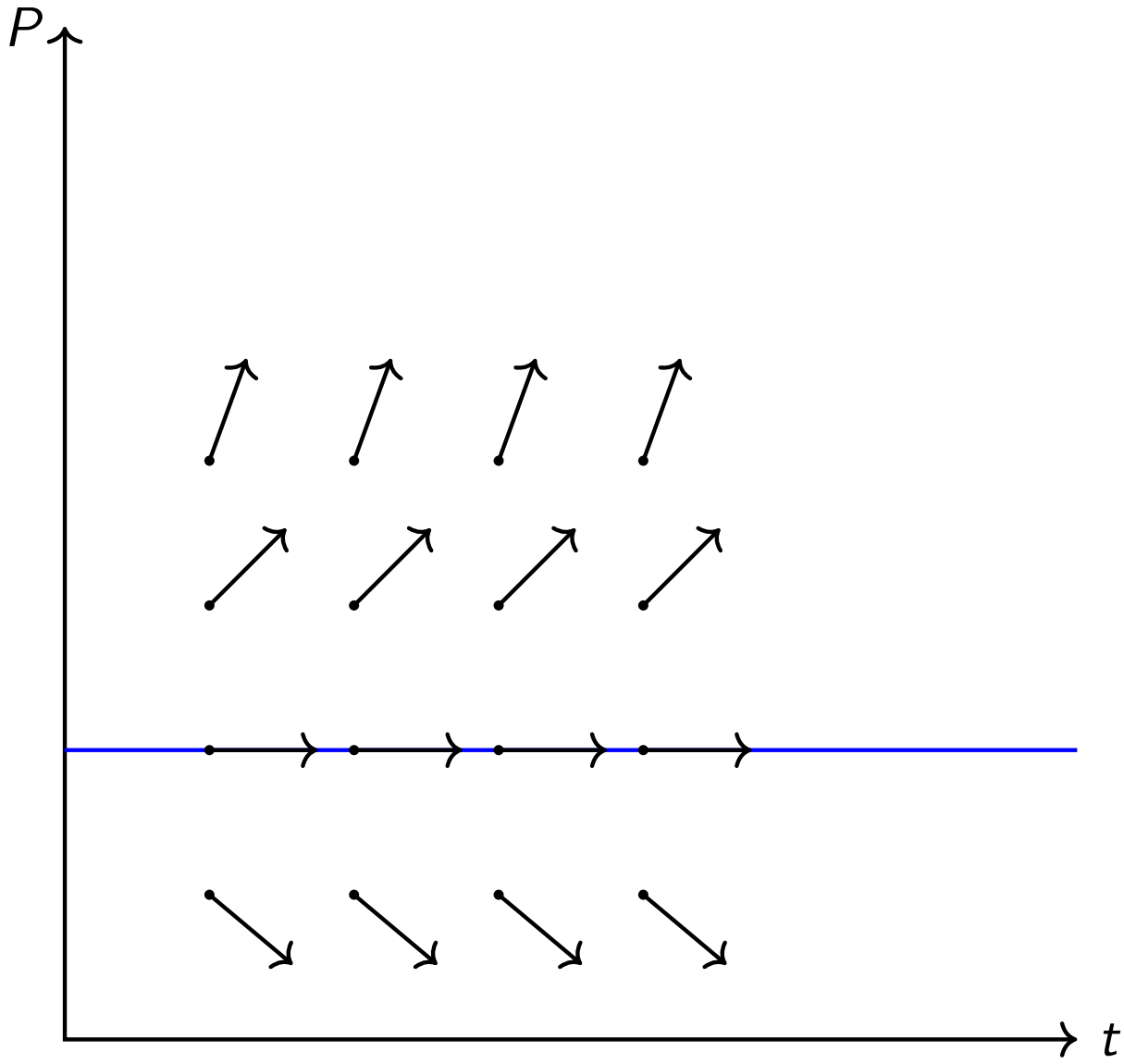
$\frac{\mathrm{d}P}{\mathrm{d}t}$ represents the rate of change or slope of the solution. We can draw the slope field or direction field on the t, P plane by evaluating the right hand side at different values of t and P.

In this example the slope is 0 for $P = 2$ so $P(t) = 2$ represents an equilibrium solution

timestamp: 30:50

Slope Field

$$\frac{dP}{dt} = 2(P-2)$$



timestamp: 31:05

Separable Equation

To solve the equation, separate:

$$\frac{\mathrm{d}P}{P-2} = 2\mathrm{d}t, \qquad \text{for } P \neq 2$$

and integrate each side

$$\int \frac{\mathrm{d}P}{P-2} = 2\int \mathrm{d}t; \qquad \ln(P-2) = 2t + A$$

where A is a constant of integration. This is an implicit solution

timestamp: 35:20

We can continue

$$\Longrightarrow \; e^{\ln(P-2)} = e^{2t+A}$$

$$\Longrightarrow \; P - 2 = \mathrm{e}^{(2t+A)} = \mathrm{e}^{2t}\,\mathrm{e}^{A} = C\,\mathrm{e}^{2t}$$

where we set $C = \mathrm{e}^{A}$, and the explicit solution is

$$P(t) = 2 + C\,\mathrm{e}^{2t}$$

timestamp: 37:07

Initial Condition

The value of C can be determined with a given initial condition,

the value of P at a fixed time (usually at $t = 0$)

Ex: if the I.C. is $P(0) = 3$ then we get $P(0) = 2 + C = 3$ so that

$C = 1$ and the solution is

$$P(t) = 2 + \mathrm{e}^{2t}$$

Initial Value Problem

An ODE paired with one or more initial conditions forms an

initial value problem (IVP)

Ex:

$$\begin{cases} \frac{\mathrm{d}P}{\mathrm{d}t} = 2P - 4 \\ P(0) = 3 \end{cases}$$

The general solution of the ODE is $P(t) = 2 + C\,\mathrm{e}^{2t}$

The specific solution of the IVP is $P(t) = 2 + \mathrm{e}^{2t}$

timestamp: 41:30